Midterm Exam

Applied Algebra — Spring Semester 2026

Stony Brook Institute at Anhui University

Instructor: Tuan Tran

Duration: 90 minutes

Name: ________________________	**Student ID #:** ________________________

Instructions:

- No notes or books are allowed. Calculators are permitted.
- Show your reasoning, not just the final answer. Answers without justification may receive only partial credit.
- If a problem asks you to solve an equation, find all solutions. If no solutions exist, explain why.

Each problem is worth **20 points**.

	1	2	3	4	5	**Total**
Grade						

Exercise 1. Circle **T** for TRUE and **F** for FALSE. If the statement is true, give a brief reason. If it is false, give a counterexample.

1. Let a, b, n be integers with $n \geq 1$. If $a \equiv b \pmod{n}$, then $a^2 \equiv b^2 \pmod{n}$. **T** **F**

   True. $a \equiv b \pmod{n}$ shows that $a = b + kn$ for some $k \in \mathbb{Z}$. Thus

   $$a^2 = b^2 + n(2bk + k^2 n) \equiv b^2 \pmod{n}.$$

2. The number $2^{20} - 1$ is divisible by 3. **T** **F**

   True. One may apply a calculator to obtain that

   $$2^{20} = 1048576 = 349525 \cdot 3 + 1.$$

3. Let a, n be integers with $n \geq 2$. Then $a^{\phi(n)} \equiv 1 \pmod{n}$. **T** **F**

   False. Take $a = n = 2026$. $a^{\phi(n)} \equiv 1 \pmod{n}$ means that $a^{\phi(n)-1}$ is the inverse of a modulo n. However, $\gcd(a, n) = 2026 > 1$, which contradicts to the existence of multiplicative inverse.

4. Define a relation R on $\mathbb{Z}$ by aRb if $a - b$ is divisible by 5. Then R is an equivalence relation. **T** **F**

   True. Check that R is reflexive, symmetric and transitive.

5. Let $f : X \to Y$ be a function. For $A \subseteq X$, define $f(A) = \{f(x) : x \in A\}$. If $f(A \cap B) = f(A) \cap f(B)$ for all $A, B \subseteq X$, then f is injective. **T** **F**

   True. For $x_1 \neq x_2 \in X$, let $A = \{x_1\}$ and $B = \{x_2\}$. Then $A \cap B = \varnothing$. Hence, $f(A) = \{f(x_1)\}$ and $f(B) = \{f(x_2)\}$ do not intersect. It follows that $f(x_1) \neq f(x_2)$, meaning that f is an injection.

Exercise 2.

(a) Solve the congruence $14x \equiv 8 \pmod{30}$. How many solutions are there modulo 30? Find all solutions.

Solution. The congruence reads that $14x = 30k + 8$, or equivalently, $7x = 15k + 4$. Hence,

$$7(x - 2k) = k + 4.$$

Thus, $k = 7\ell + 3$, and $x = 15\ell + 7$ for some $\ell \in \mathbb{Z}$. In conclusion, there are two solutions modulo 30: 7 and 22. □

(b) Find all solutions modulo 45 of the system

$$x \equiv 2 \pmod{5}, \qquad x \equiv 3 \pmod{9}.$$

Determine the smallest nonnegative solution.

Solution. Notice that $\gcd(5, 9) = 1$. Chinese remainder theorem guarantees that the solution modulo $5 \cdot 9 = 45$ is unique. Check that $x = 12$ is indeed a solution, which is also the smallest nonnegative one. In conclusion, $x \equiv 12 \pmod{45}$. □

Exercise 3. The Fibonacci numbers are defined by $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$.

(a) Show that $F_n \equiv F_{n-3} \pmod 2$ for every $n \geq 3$.

Proof. By definition,

$$F_n = F_{n-1} + F_{n-2} = 2F_{n-2} + F_{n-3} \equiv F_{n-3} \pmod 2.$$

□

(b) Show that F_n is even if and only if n is divisible by 3.

Proof. By (a), if $n \equiv m \pmod 3$, then $F_n \equiv F_m \pmod 2$. It follows that

$$F_{3k+2} \equiv F_2 \equiv 1 \pmod 2; \; F_{3k+1} \equiv F_1 \equiv 1 \pmod 2; \; F_{3k} \equiv F_0 \equiv 0 \pmod 2.$$

□

Exercise 4. Let n be a positive odd integer. In this exercise, we will prove that $n \mid 2^{(n-1)!}-1$.

(a) Let p^r be a prime power divisor of n. Show that $\phi(p^r)$ divides $(n-1)!$.

Proof. Since $p \neq 2$ is a prime number, $\phi(p^r) = p^{r-1}(p-1)$. Observe that

$$(n-1)! = (n-1)(n-2)\cdots 1,$$

and

$$1 \leq p-1 \neq p^{r-1} < p^r \leq n.$$

Then $p-1$ and p^{r-1} both appear in the expansion of $(n-1)!$. In conclusion, $\phi(p^r)$ divides $(n-1)!$. $\square$

(b) For such a prime power p^r, show that $2^{(n-1)!} \equiv 1 \pmod{p^r}$.

Proof. $\gcd(2, p^r) \leq \gcd(2, n) = 1$. By (a), $(n-1)! = \ell \cdot p^r$ for some $\ell \in \mathbb{N}$. Euler's theorem shows that

$$2^{(n-1)!} = \left(2^{\phi(n^r)}\right)^{\ell} \equiv 1 \pmod{p^r}.$$

$\square$

(c) Deduce that $2^{(n-1)!} \equiv 1 \pmod{n}$.

Proof. By definition, $\phi(n) \leq n-1$. Hence $\phi(n)$ divides $(n-1)!$. Let $(n-1)! = k \cdot \phi(n)$. Notice that n is odd, or $\gcd(2, n) = 1$. Euler's theorem then shows that

$$2^{(n-1)!} = \left(2^{k}\right)^{\phi(n)} \equiv 1 \pmod{n}.$$

$\square$

Exercise 5. Consider an automaton with alphabet $\{a, b\}$ and two states S_0 and S_1. The initial state is S_0. The transition rules are as follows: from S_0, reading a stays in S_0 and reading b goes to S_1. From S_1, reading a stays in S_1 and reading b goes to S_0. The accepting state is S_0.

(a) Starting from S_0, list the sequence of states visited while reading $abba$.

Solution.

$$S_0 \xrightarrow{a} S_0 \xrightarrow{b} S_1 \xrightarrow{b} S_0 \xrightarrow{a} S_0.$$

□

(b) Does the machine accept the string $abab$?

Solution.

$$S_0 \xrightarrow{a} S_0 \xrightarrow{b} S_1 \xrightarrow{a} S_1 \xrightarrow{b} S_0.$$

Thus, the machine accept this string. □

(c) Give two different strings of length 2 that are accepted.

Solution. Check by definition that aa and bb are accepted. □

(d) Characterize all strings that are accepted.

Solution. By the transition rules, reading a does not change while reading b switches between S_0 and S_1. Since the initial state and the accepting state are both S_0, a string is accepted if and only if it has even number of 'b'. □